

Representation of Geographical Features

i) keywords:

- | | |
|----------------------|--------------------|
| 1) Topographical Map | 6) Arbitrary |
| 2) Cultural | 7) Map distance |
| 3) Physical | 8) Ground distance |
| 4) Scale | 9) Contours |
| 5) Grid | 10) Townlines |
| | 11) Swells |

ii) Teaching Points

- * Topographical maps - The survey of India makes large scale maps based on actual survey of land, these maps are called topographical maps or topo sheets.
- * Grid - The network of parallels and meridians on the map drawn in the black colour to the correct location to the place on the map.
- * Arbitrary grid system - A set of vertical &

horizontal lines drawn in red colour on a Toposheet to locate places.

Topographical Map

* Survey of India

* Scales - 1 : 25,000

* Grid

* Arbitrary grid system

* Uses of colour on topo sheet

• Blue - water bodies, seas, ocean, rivers, ponds, lakes etc.

• Red - settlements, includes roads, tracks, and arbitrary grid system.

• Yellow - cultivated areas

• Brown - Contours, farm lines, sand dunes, sand hills

• Green - forested areas, jungles, scrub.

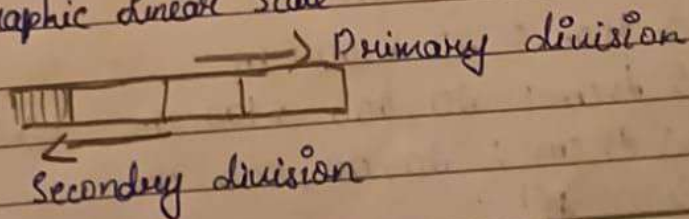
- Black - abbreviations, surveyed trees, railway tracks, telephone and telegraph lines, etc
- White - Barren land or uncultivated area

* Scales

- In India, we use metric system to measure the distance like cm, m, km
- Map distance
- Ground distance
- Scale.

* Types of Scales

- By a Statement Scale - $1 \text{ cm} : 1 \text{ km}$
- By a Representative fraction - $\frac{1}{1,00,000}$
- By a Graphic Linear Scale



Numericals -

3. If the RF is $\frac{1}{10,00,000}$, find the statement scale

Ans $\Rightarrow RF = \frac{1}{10,00,000}$

$\Rightarrow 1\text{cm} : 10,00,000 \div 1000$

$\Rightarrow 1\text{cm} : 10,00,000 \div 1000$

$\Rightarrow 1\text{cm} : 1000 \div 100$

$\Rightarrow 1\text{cm} : 10\text{ Km}$

~~HN - I~~ [D'main, Q1 and 2]

Q1. Distinguish between a grid and an arbitrary grid system

Grid	Arbitrary grid system
The network of parallels and meridians on the map drawn in black colour to the current location of a place on the north	A Set of vertical & horizontal lines drawn in red colour on toposheet to locate places

Q2. Write the names of three features that brown and green each show on a topographical map.

Ans. Brown - contours, farm lines, sand dunes, sand wells.

• Green - forested areas, jungles, scrubs, orchards.

Q4. If the R.F. is $\frac{1}{4,00,000}$, find the Statement Scale.

Ans $\Rightarrow \text{RF} = \frac{1}{4,00,000}$

$\Rightarrow 1 \text{ cm} = 4,00,000 \text{ cm}$

$\Rightarrow 1 \text{ cm} = 4,00,000 \div 1000$

$\Rightarrow 1 \text{ cm} = 400 \div 100$

$\Rightarrow 1 \text{ cm} = 4 \text{ km}$

Q1. List the steps to measure the distance of straight line on a map.

Ans. Tools - scale, strip of paper, divider.

- 1) First make the points, say A and B, which the distance is to be measured.
- 2) Place the strip of edge of paper along the line AB.
- 3) Mark the limits of distance on the paper.
- 4) Now place the paper along the linear scale and read the map distance.
- 5) The distance between points A and B can also be measured.

Q2. List the steps to measure the distance of a curved line on a map.

Ans Tools - Thread, String, Scale, opisometer.

- 1) First mark the points, say A and B, between which the distance is to be measured.
- 2) Place one end of the thread on point A and then move the thread along the curved course.
- 3) Now place the thread along the linear scale & read the map distance.
- 4) An instrument, called opisometer, can also be used to measure the correct map distance.

ground distance indicated on its dial.

Conventional & Symbols

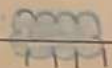
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- A map does not have enough space to accomodate actual size and shape of various physical and cultural features, but a map without these features is less informative. So, there are signs, symbols and letters are used to show various information on the map. These are called conventional signs & symbols.

Conventional signs and symbols.

Signs and Symbols	Meaning
1	Metalled Road
2	Unmetalled road
3	Unmetalled road with bridge
4	Cart track
5	Dank track
6	Cause way
7	Dry Stream
8	Perennial river
9	Dry stream with water canal.

Conventional signs and symbol

	Signs and symbols	Meaning
10.		River with Island
11.		Tidal water
12.		Lined wall
13.		Unlined wall
14.		Perennial Tank
15.		Dry tank
16.		Contours
17.		Sand hills and sand dunes
18.		Towns and villages
		Permanent hut.

Conventional signs and symbols.

	Signs and symbols	Meaning
20		Temporary hut
21		Temple.
22		Chattri
23	Δ 200	Triangulated height
24	390	Spot height
25	BM 63.3	Bench mark
26		Post office
27		Police station
28	P.T.O	Post and telegraph office
29	50	Telegraph office

HW-2 (D main)

3. How are the scales stated in the statement representative fraction and graphic method.

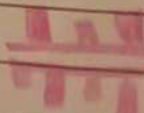
Ans * Statement - Scale is expressed in 2 words.
1 cm : 5 km. In this statement 1 cm is the map distance and 5 km is the ground Distance.

* Representative fraction - Scale is expressed in numerical fraction. In this period / method the ratio between the map distance and ground distance is given as fraction

* Graphic Scale - Map distance is shown as a straight line.

5. Draw conventional signs and symbols for:

• Footpath with bridge. 

• Inhabited town 

• Height: 1000, 2000

• Palomura:

• Tidal river: 

Terms to remember

1. Cultural features - features made by humans such as roads, forts, bridges, and so on.
2. Contour - a line drawn on a map which join places and have the same height above the mean sea level.
3. faunline :: approximate contour lines
4. scuabs - vegetations containing stunted trees or shrubs.

Conventional signs and symbols

Signs and symbols

Meaning

DB

Dak bungalows

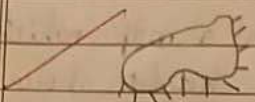
IB

Inspection bungalows

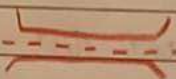
CH

Circuit house

District Boundary



Embankment tank



Footpath with bridge



Broken ground

II. Answer the given question

1. Why is the statement scale not reliable when the map is photocopied or enlarged?

ans. The statement scale is not reliable when the map is photocopied or enlarged as:

- * The map distance on the map changes its size.
- * It alters ratio between map distance and ground distance.

2. Why are conventional signs and symbols important?

ans. * The conventional signs have space on the map and helps to accommodate many features showing their actual size and exact shape.

* It makes it easier to represent physical and cultural features and it ensures universal understanding between the limited area.

3. Why are settlements often located close to a river or perennial river?

ans. * Availability of water: river provides water for agriculture, household use, rearing livestock.

* Fertile land: It provides fertile alluvial soil.

* Transport and connecting: water transport

4. Differentiate between map and ground distance.

ans.

Map distance

Ground Distance

• Distance between any two points on a map measured in a straight line.

• Distance between same two points on ground measured in a straight line.

5. Describe the importance of scale in reading a topo sheet.

ans * The scale determines the actual ground distance for the places shown on the map

* It allows the user to understand the level of details provided in large-scale maps for smaller area.

* It ensures the management for the relief features, transport, routes, settlements spacing with complete details interpreted correctly.

6. What information can contour lines provide on a topo sheet.

ans * It helps us to elevations, relief features and to identify whether they are hills, ridges, valleys and plateaus.

* The spacing of the contour lines indicates whether the mountain is steep or gradually steep.

Close contour lines : Steep slopes
Wide spaced contour lines : gradual slopes

* The contour patterns helps us to identify specific landforms like conical peaks, hills, valley's etc.

Hw-3 (c main)

1. On a topo sheet, yellow and green colour show desert and plain". Is this statement correct? Give reason.

Ans No, as yellow is used to show cultivated area and green is used to show forested and wooded areas, orchards, trees and scrubs.

2. Find the Statement of Scale in km of a topo sheet whose R.F is $\frac{1}{50,000}$

$$\Rightarrow R.F = \frac{1}{50,000}$$

$$1 \text{ cm} = 50,000 \text{ cm}$$

$$1 \text{ cm} = 50,000 \div 1000 \text{ m}$$

$$1 \text{ cm} = 50 \div 100 \text{ Km}$$

= 1 cm = 0.5 Km

= 2 cm = 1 Km.